MATH 127 Calculus for the Sciences

Lecture 28

Today's lecture

Last time
Curve sketching

This time
Course note coverage Section 3.4
More curve sketching

Rant about the sign table thing

Example Suppose you are doing some question, and at one point you have computed $f'(x)$ to be

$$(x^2 - 4)\ln(x)$$

from which found that the critical points are $x = 1$ or $x = 2$.
In the next step, you want to find the signs of $f'(x)$ and use it to determine intervals of increase/decrease, or to determine whether those critical points are local max or min or neither. What do you do to find the signs?

Remember those tables of signs I did, which you also see in the course notes?

To figure out what signs to put in each cell of the table, you need to do some inequalities, and I'm bad at inequalities. Let me propose a better method.

About the sign table thing

Example Find the signs of

$$(x^2 - 4)\ln(x)$$

knowing it is equal to zero at $x = 1$ or $x = 2$.

1. Draw the x-axis, and split it using the critical points. (In this example, we only draw the positive half, because $(0, \infty)$ is the $\boxed{}$).

2. On each piece of this splitting, pick any value and plug it into $f'(x)$. Ideally you pick a nice number that is easy to compute, or easy to treat conceptually
   2.1 On the interval $(0, 1)$, we have

$$f'(0.00001) = \boxed{} \times \boxed{} = \boxed{}$$

   On the interval $(1, 2)$, we have

$$f'(1.99999) = \boxed{} \times \boxed{} = \boxed{}$$

   On the interval $(2, \infty)$, we have

$$f'(e^{999}) = \boxed{} \times \ln(e^{999}) = \boxed{}$$

3. Just label each piece with the sign of your answer and you're done.

About the sign table thing

Example Suppose $f''(x)$ is

$$3x(x+1)^2$$

and we have inflection points at $x = 0$ and $x = -1$. Find intervals concavity.

Curve sketching

Steps to curve sketching

1. Domain.
2. Obvious points: intersection with x or y axis; other easy-to-compute points.
3. Horizontal asymptotes: find limits at $x \to \pm\infty$
4. Vertical asymptotes: list out potential points, then find left/right limits at those points.
5. Critical point: find when $f'(x) = 0$ *or undefined*.
6. Interval of increase/decrease: find signs of $f'(x)$ like what we did before.
7. Inflection points: find when $f''(x) = 0$.
8. Interval of concavity: find signs of $f''(x)$ like what we did before.

Example

Give me some examples you want me to do.
Candidates:

- $xe^{-1/x}$
- $\sin\left(\frac{1}{x}\right)$
- $\tan\left(x^2\right)$
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